

B.C.A. (Honours) & B.C.A. (Honours with Research)
(Semester - 3 and Semester - 4)
Saurashtra University
To be effective from June – 2024

BCA-4	
CS-24 Operating system concepts with Unix/Linux	
Minimum following exercise should be performed by the students during the semester	
(1)	Basic Linux Commands Learn and execute simple file & directory commands — pwd, ls, cd, mkdir, rmdir, rm, cp, mv, cat, touch. (Folder create, move, copy, delete, show path, etc.)
(2)	File Permissions Change file access rights and ownership using chmod, chown, chgrp. (Read, Write, Execute permission set karva.)
(3)	Viewing & Filtering Files Display and count file content using head, tail, wc, sort, uniq. (First/last lines, sorting, removing duplicates.)
(4)	Searching & Pattern Matching Search specific text or patterns using grep, egrep, fgrep.
(5)	Cut, Paste & Translate Extract specific columns, merge files and replace characters using cut, paste, tr.
(6)	Use of tee Command Show output on screen and save to a file at the same time using pipes and tee.
(7)	File Compression & Archiving Create and extract compressed archives using tar, gzip, gunzip, zip, unzip.
(8)	Process Management Check and control running processes using ps, top, kill, jobs, bg, fg, nice, renice.
(9)	Text Editing with VI/Nano Create and edit files, insert, delete, search, save and quit using VI or Nano editors.
(10)	Basic Shell Script Write a simple script using variables, echo, and read.
(11)	Decision Making Scripts Use conditions in shell scripts — if, if-else, case statements.
(12)	Looping in Shell Scripts Repeat tasks using for, while, and until loops.
(13)	Command Line Arguments Write shell scripts that accept and use command line arguments (\$1, \$2...).
(14)	Functions in Shell Scripts Create and use user-defined functions for reusable code.
(15)	To perform demonstration of WINE.
(16)	User & Group Management Create, delete, and manage users & groups — useradd, passwd, who, w, id.
(17)	Job Scheduling Schedule and automate tasks using at, batch, cron, crontab.
(18)	Searching Files in System Locate files and commands using find.
(19)	Backup & Restore Create and restore backups using tar and gzip.